SOFTWARE REQUIREMENTS SPECIFICATION

Content Management System (CMS)

PHP and MySQL

Document Item	Value
Document Type	Software Requirements Specification (SRS)
Project	Content Management System (CMS)
Academic Context	BCA – 5th Semester Mini Project
Primary Technology	PHP
Database	MySQL
Document Status	Project Baseline
Version	1.0
Prepared For	Academic submission and implementation reference

Problem Statement

A blog site requires a CMS that allows authors to write, edit, and publish articles. The system shall be implemented using PHP and MySQL, allow authors to log in, create new posts, edit existing posts, and delete posts, and display posts to the public in a structured format.

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Note: Page numbers are intentionally omitted because this document is intended to remain editable during implementation.

1. Introduction

1.1 Purpose

This document defines the software requirements for a Content Management System (CMS) for a blog website. The purpose of the system is to give authorized authors a simple interface for managing article content while providing the public with a consistent and structured way to read published articles.

1.2 Problem Statement

The project is based on a blog site that needs centralized content management. Without a dedicated CMS, article creation, editing, publishing and removal can depend on manual handling of website content. The proposed system provides these operations through a PHP and MySQL application.

1.3 Objectives

- Provide a secure author login mechanism.
- Allow authenticated authors to create new blog posts.
- Allow authors to edit existing posts.
- Allow authors to publish articles when they are ready for public viewing.
- Allow authors to delete posts they are permitted to manage.
- Display published posts publicly in a clear and structured format.
- Maintain article data in a MySQL database rather than hard-coding posts into web pages.

1.4 Intended Audience

The document is intended for the student developer, project guide or faculty reviewer, testers, and anyone responsible for maintaining or evaluating the CMS.

2. Overall Description

2.1 Product Perspective

The CMS is a web-based application consisting of a server-side PHP layer, a MySQL database, and web pages used by authors and public visitors. Author functions are protected by authentication. Public users can browse published articles without accessing author management functions.

2.2 Product Functions

Area	Core Function
Authentication	Author login and logout
Content Creation	Create a new article
Content Editing	Edit an existing article
Publishing	Make an article visible to the public
Deletion	Remove an article from the CMS
Public Website	List and display published articles
Storage	Persist users and posts in MySQL

2.3 User Classes

User	Responsibilities
Author	Logs in and manages blog posts through the CMS.
Public Visitor	Views published articles and navigates the public blog.

2.4 Operating Environment

The system is intended to run as a PHP web application connected to a MySQL database. For academic development and testing, a local Apache/PHP/MySQL environment such as XAMPP may be used. The browser-facing portion should work on current desktop browsers.

3. System Scope

3.1 In Scope

- Author authentication.
- Creation, viewing, editing, publishing and deletion of posts.
- Storage of post and author information in MySQL.
- Public display of published posts.
- Protection of author-only functions from unauthenticated visitors.

3.2 Out of Scope for Version 1.0

- Comments and discussion features.
- Likes, followers, social networking functions and private messaging.
- Online payments or subscription billing.

- Multi-organization tenancy.
- Advanced workflow with separate editor and administrator roles.
- Mobile application.

These items may be considered later without changing the core purpose of the CMS.

4. Functional Requirements

ID	Requirement	Description	Priority
FR-01	Author authentication	The system shall provide a login form through which a registered author can submit valid credentials.	High
FR-02	Login validation	The system shall validate submitted credentials and deny access when the credentials are invalid.	High
FR-03	Session management	The system shall maintain an authenticated author session until logout or session expiry.	High
FR-04	Logout	The system shall provide a logout operation that terminates the author session.	High
FR-05	Create post	An authenticated author shall be able to create a new post by entering the required article information.	High
FR-06	Store post	The system shall save a newly created post in the MySQL database and associate it with the author who created it.	High
FR-07	Edit post	An authenticated author shall be able to edit an existing post that they are permitted to manage.	High
FR-08	Update post	The system shall persist edited post information in MySQL without creating an unrelated duplicate post.	High
FR-09	Publish post	An authenticated author shall be able to mark a post as published.	High
FR-10	Public visibility	Only posts marked as published shall be shown in the public blog area.	High
FR-11	Delete post	An authenticated author shall be able to delete a post they are permitted to manage.	High
FR-12	Deletion confirmation	The system should request confirmation before completing an irreversible delete action.	Medium
FR-13	Public post listing	The system shall display published posts in a structured list containing, at minimum, the post title and basic article information.	High
FR-14	Article view	The system shall provide a public view for reading the full content of a published post.	High
FR-15	Database relationship	The system shall maintain a relationship between an author record and the posts created by that author.	High
FR-16	Access control	The system shall prevent unauthenticated users from accessing author management pages.	High
FR-17	Validation	The system shall validate required post fields before saving or updating a post.	High
FR-18	Output safety	The system shall safely render stored text so that submitted HTML or script content cannot execute as active client-side code when it is not explicitly supported by the application.	High
FR-19	Error handling	The system shall display a meaningful error message when a database or application operation cannot be completed.	Medium
FR-20	Post ordering	The public post listing should display recent published posts before older published posts.	Medium

5. Use Cases

UC-01 Author Login

Item	Description
Primary Actor	Author
Precondition	The author has a valid account.

Item	Description
Main Flow	Author opens login page → enters username/email and password → submits form → system validates credentials → system creates session → dashboard is displayed.
Alternate Flow	Invalid credentials → system rejects login and displays an error.
Postcondition	Author is authenticated.

UC-02 Create and Publish Post

Item	Description
Primary Actor	Author
Precondition	Author is logged in.
Main Flow	Author opens create-post page → enters title and content → saves post → system stores post → author chooses publish → post becomes publicly visible.
Postcondition	The post exists in the database and is publicly visible when published.

UC-03 Edit Post

Item	Description
Primary Actor	Author
Precondition	Author is logged in and has permission to manage the post.
Main Flow	Author opens the post from the dashboard → changes the required fields → submits update → system validates and saves changes.
Postcondition	The stored post reflects the latest approved edit.

UC-04 Delete Post

Item	Description
Primary Actor	Author
Precondition	Author is logged in and has permission to manage the post.
Main Flow	Author selects delete → system requests confirmation → author confirms → system removes the post → dashboard is refreshed.
Postcondition	The post is no longer available through the CMS and is not displayed publicly.

UC-05 View Published Article

Item	Description
Primary Actor	Public Visitor
Precondition	A published article exists.
Main Flow	Visitor opens the blog → selects an article → system retrieves the published post → full article is displayed.
Postcondition	Visitor can read the article without author authentication.

6. Data Requirements

6.1 Author Data

Field	Purpose	Example / Type
id	Unique author identifier	INT, primary key
username or email	Login identifier	VARCHAR
password_hash	Stored password representation	VARCHAR
created_at	Account creation timestamp	DATETIME/TIMESTAMP

6.2 Post Data

Field	Purpose	Example / Type
id	Unique post identifier	INT, primary key
author_id	Identifies the author of the post	INT, foreign key
title	Article title	VARCHAR
content	Article body	TEXT
status	Controls public visibility	ENUM or equivalent
created_at	Creation timestamp	DATETIME/TIMESTAMP
updated_at	Last modification timestamp	DATETIME/TIMESTAMP

6.3 Logical Relationship

One author may create multiple posts. Each post belongs to one author.

AUTHOR (1) ——————— (N) POST

7. External Interface Requirements

7.1 Author Interface

- Login page with credential fields and validation messages.
- Author dashboard showing the author's posts and their current status.
- Create/Edit Post form containing required article fields and a publish action.
- Delete action with a confirmation step.

7.2 Public Interface

- Public homepage or blog listing.
- Structured article cards or entries.
- Full article page for each published post.
- Navigation that does not expose author-management functions to public visitors.

7.3 Database Interface

PHP shall communicate with MySQL to create, read, update and delete author and post records. Database operations should use parameterized/prepared statements where supported.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	Passwords shall not be stored as plain text. Password verification shall use a secure password hashing mechanism.
NFR-02	Security	Database inputs shall be handled using prepared/parameterized statements to reduce SQL injection risk.
NFR-03	Security	Author-only pages shall require an authenticated session.
NFR-04	Security	User-controlled output shall be safely escaped or otherwise sanitized before being rendered as HTML.
NFR-05	Usability	Common author actions such as creating, editing and publishing a post should be understandable without technical knowledge.
NFR-06	Performance	Normal page requests should return without noticeable delay under the expected load of a small blog.

ID	Category	Requirement
NFR-07	Reliability	A failed database operation shall not leave the application in an unclear state; the user should receive an error message.
NFR-08	Maintainability	PHP code should be organized into separate responsibilities such as configuration, authentication, post management and public display.
NFR-09	Compatibility	The public site should render correctly on current desktop web browsers.
NFR-10	Data Integrity	Foreign-key relationships and validation rules should prevent orphaned or invalid post records.

9. Business Rules and Constraints

- Only authenticated authors may access post-management functions.
- A post must have the required fields before it can be saved.
- A post is publicly visible only when its status is set to published.
- An author may edit or delete only posts that the application permits that author to manage.
- Deleting a post is an irreversible operation in Version 1.0 unless a future recycle-bin feature is introduced.
- MySQL is the system of record for users and posts.
- The implementation shall use PHP for server-side processing and MySQL for persistent storage.

10. Acceptance Criteria

1. Given valid author credentials, the author can log in and reach the CMS dashboard.

2. An authenticated author can create a post and the post is stored in MySQL.

3. An authenticated author can edit an existing post and the changes are stored correctly.

4. An authenticated author can publish a post and the public site then displays it.

5. An unpublished post is not shown on the public site.

6. An authorized author can delete a post after confirmation.

7. An unauthenticated visitor cannot access author-only management pages.

8. The public site displays published posts in a structured, readable format.

9. Invalid input or database failures produce an understandable error message.

10. Passwords are not stored as plain text and database write/read operations use safe query handling.

11. Requirements Traceability

The following matrix shows how the core problem statement maps to the system requirements.

PS Requirement	Mapped SRS Requirements
Authors can log in	FR-01, FR-02, FR-03, FR-04, FR-16
Authors can write/create articles	FR-05, FR-06, FR-17
Authors can edit existing posts	FR-07, FR-08, FR-17
Authors can publish articles	FR-09, FR-10
Authors can delete posts	FR-11, FR-12
Public can see posts in structured format	FR-13, FR-14, FR-20
PHP and MySQL implementation	Section 2.4, Section 7.3, Business Rules

12. Future Enhancements

The following features are intentionally outside the Version 1.0 baseline. They can be added later without changing the core CMS architecture.

- Categories and tags for improved article organization.
- Keyword search and category-based filtering.
- Draft autosave and revision history.
- Separate Editor and Administrator roles.
- Scheduled publication and automatic unpublishing.
- Featured articles and homepage controls.
- Image/media library.
- Article analytics such as views.
- Comment moderation.
- Email notifications for publication or review actions.